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Decompression of magma into repository tunnels

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It is nontrivial to find and design safe repository sites for nuclear waste. It appears common sense to drill tunnels as repository sites in a mountain in remote and relatively dry regions. However, erosion of the waste canisters by naturally abundant chemicals in the mountains water cycle remains a problem. Other problems may be caused by geophysical phenomenon like volcanic eruptions. Yucca Mountain is a location which has been proposed as storage site. We investigate the consequences of evolving magma flows in the repository systems, because the probability of magmatic activity at Yucca Mountain is too high to neglect an investigation of volcanic effects.

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AMS Subject Classification: 93A30, 35L65, 35L15, 74A50

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1 Yucca Mountain

The United States face the difficult task to decide whether Yucca Mountain (YM) in the state Nevada is a safe site to store nuclear waste for the following 10.000 years [1,2]. YM is situated in a landscape formed by volcanic activity during the last one million years (Fig. 1). Given these volcanic remains, geophysicists have estimated the probability for future eruptions through YM to be about 10^{-7} per year, or 1 : 1000 over the 10.000 year storage period. This probability lies around the threshold for which the possible effects of catastrophic events, such as volcanic eruptions, on nuclear facilities such as reactors and storages sites need to be investigated in detail and included in the overall risk analysis. Hence, we [2] were asked by the U.S. Nuclear Regulatory Commission to provide a first description of the flow evolution through the repository system following a volcanic eruption. The NRC is an independent commission which reviews whether the research of the Department of Energy into the safety of the YM repository site is adequate.

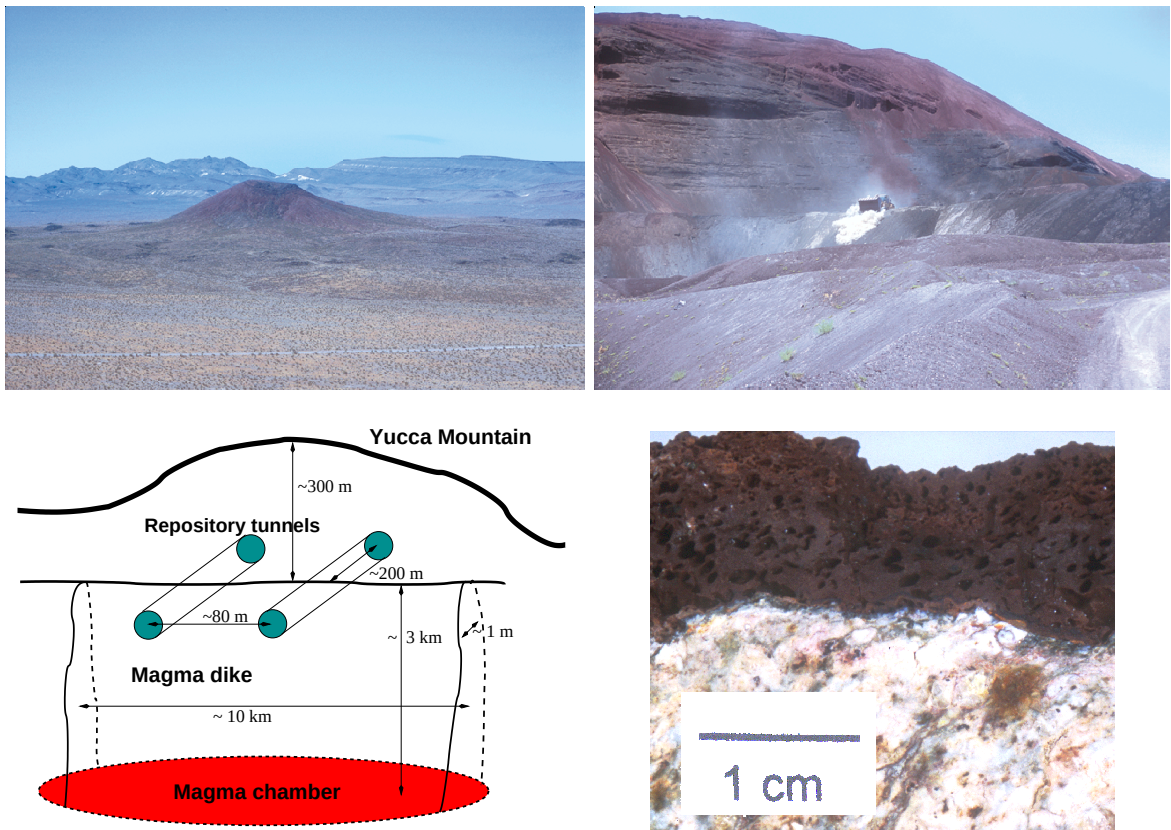


Figure 1: Red Cone was formed about one million years ago. The foothills of YM are seen in the background on the right. Lathrop Wells is about 100.000 years old, and her cinder cone is mined. Below left is a sketch of a magma dike intersecting the repository tunnels of YM. The layer of magma on the piece of ignimbrite lining of the conduit below Lathrop Wells shows, below right, the characteristic holes the water vapor bubbles have left behind. (Photo's: zw/artprojects.)

2 “Wet” basaltic magma with water vapor bubbles

The magma below YM consists of basalt with $n_0 = 1\text{--}2\text{ wt}\%$ (weight percent) water. At a depth of a few kilometers the water is dissolved in the magma that has a constant temperature around $T = 1350\text{ K}$. The density of this nearly incompressible mixture lies around $\sigma = 2500\text{ kg/m}^3$. Under its own weight the pressure in the magma increases as function of depth.

The magma moves as a dike of width $1\text{--}2\text{ m}$ and a few kilometers length upward through the rock at a speed of $1\text{--}2\text{ m/s}$ as a consequence of an excess pressure in the magma chamber (at $3\text{--}30\text{ km}$ depth) above the hydrostatic pressure. Depending on the magnitude of this excess pressure in the magma chamber, the magma dike reaches the Earth’s surface or it stops below this surface. The repository tunnels lie at a depth of about $h = 300\text{ m}$ at which the pressure at the dike tip is about the hydrostatic pressure, $p_h = \sigma g h = 7.5\text{ MPa}$, plus an excess pressure in the magma chamber of about $0\text{--}10\text{ MPa}$. Water vapor bubbles appear when magma decompresses as it approaches the surface.

Finally, as a consequence, there will form an effectively compressible multiphase fluid-gas mixture which can lead to a strong, explosive decompression if this mixture suddenly encounters air at atmospheric pressure. In nature, these explosive eruptions may occur by breaking a magma plug in a crater of a volcano, for example, due to a landslide when a critical pressure is reached [3]. A simplified, macroscopic equation of state of this multiphase mixture gives the density, ρ , as function of pressure p (see Appendix A, (3.1) or [5]).

3 Flow tube model

We can *in principle* calculate the flow of magma and air in the dike and tunnel by using this simplified equation of state (3.1), and the equations of motion expressing mass and momentum balance of magma and air. However, these 3D equations are too complicated. We will therefore consider the fluid dynamics in a so-called flow tube model consisting of the segment of the magma dike which feeds one of a series of tunnels, and the tunnel itself (see Fig. 2). We average the flow over the cross section $A(\xi_1, t)$ of the flow tube. The coordinate ξ_1 along the flow tube is thus vertical in the magma dike, horizontal in the tunnel, and has as smooth connecting part which shape we estimate in between. We can also derive the equations of motion of the magma and air directly (see Appendix A, (3.2) or [4,5]). The final result is a reasonably straightforward mathematical model describing the density ρ , momentum ρu and velocity u along the tube as function of ξ_1 and time t .

Equation of state and mass conservation

The mass fraction $n(p)$ of water vapor in the magma has been measured in the laboratory by petrologists at constant temperature as function of pressure p , $n(p) = n_0 - s_H \sqrt{p}$, in which n_0 is the total mass fraction of water and $s_H \approx 3 \times 10^{-6}\text{ Pa}^{-1/2}$ is Henry’s constant. The simplified equation of state is:

$$\rho(p) = \left(\frac{n(p) R_v T}{p} + \frac{1 - n(p)}{\sigma} \right)^{-1} \quad (3.1)$$

with $R_v = 462\text{ J kg}^{-1}\text{ K}^{-1}$ the gas constant for water vapor. Note that $n(p) = 0$ implies $\rho = \sigma$: at high pressure the magma is free of bubbles, and that $R_v T/p$ equals the density

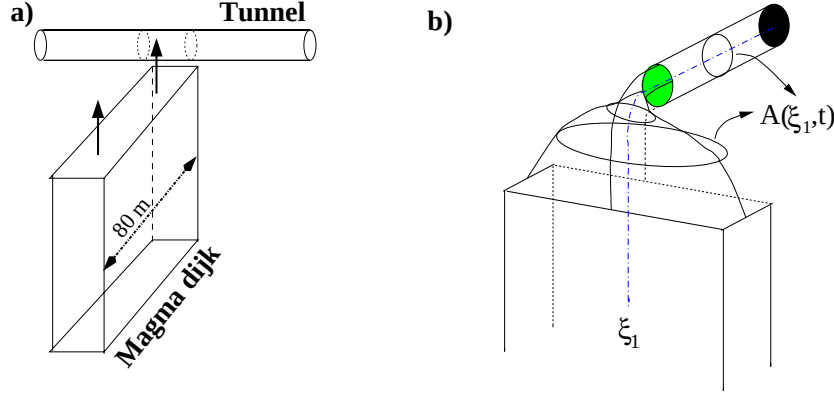


Figure 2: (a) A magma dike moves upward through the rock and either reaches the surface or a subsurface repository or mining tunnel. (b) In a flow tube model the magma dike and the tunnel are smoothly coupled and the flow is averaged over a (reasonable) cross section $A(\xi_1, t)$. The configuration is chosen to be symmetric: only the right part of the flow is considered. At both ends the tunnel is closed.

of the water vapor according to the ideal gas law.

The mass conservation equation in the flow tube emerges by equating the in- and out flow of mass in a slice of magma with thickness $\Delta\xi_1$ and cross section $A(\xi_1, t)$ with the change of mass of magma in the segment as function of time:

$$\frac{\partial(\rho A)}{\partial t} \Delta\xi_1 = -(\rho A u)|_{\xi_1+\Delta\xi_1} + (\rho A u)|_{\xi_1} \implies \frac{\partial(\rho A)}{\partial t} + \frac{\partial}{\partial\xi_1}(\rho A u) = 0. \quad (3.2)$$

We can derive the other equations similarly, whereafter we also add frictional effects.

4 Rapid decompression

When at $t = 0$ the magma dike intersects the tunnel we expect that the magma will rapidly enter the tunnel due to the large pressure difference. After some geometrical simplifications, we can already make a first analytical prediction of the pressure profile as function of space and time by solving the so called shock tube problem [4,5]. In an experimental shock tube set-up the pressure of the two gases on either side of a diaphragm is initially different and the flow is at rest. After removing the diaphragm a discontinuity in pressure and density or a shock wave will travel into the low-pressure gas region and a decompression wave into the high-pressure gas region, while the magma-air interface will follow and lag behind the shock. Without these geometrical simplifications we will have to resort to numerical simulations in order to determine pressure profiles, see Fig. 3 for pressure profiles for two different cross sections $A(\xi, t)$. In the first pressure profile at $t = 0$ for either case we clearly notice that the pressure in the last 200 m ($\xi_1 \sim \in [800, 1000] \text{ m}$), in the tunnel filled with air, is low, at atmospheric pressure, while the dike is filled with high pressure magma. Initially, neither the magma nor the air flows.

In the subsequent pressure profiles, shock waves develop. The tunnels are seen to get

filled in 10 – 120 s depending on the initial dike-tip pressure and $A(\xi_1, t)$. The final pressure roughly equals this initial dike tip pressure.

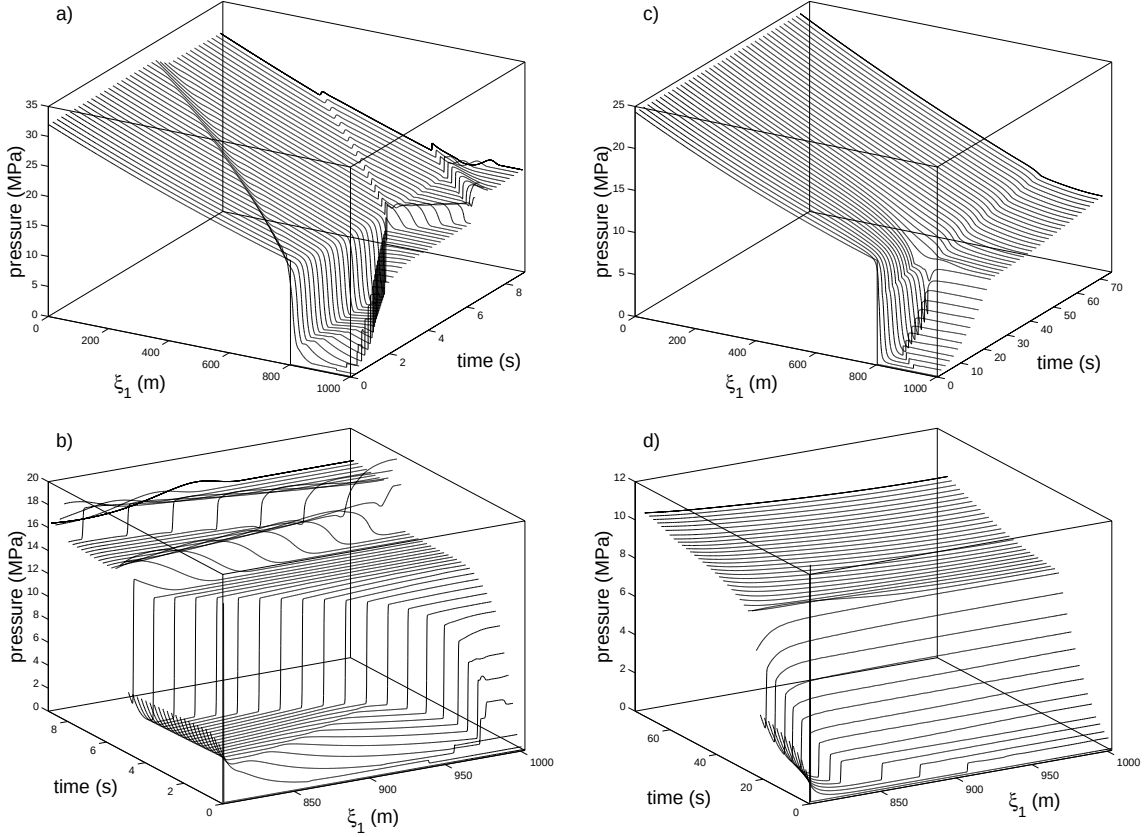


Figure 3: a,b) The pressure profiles in the dike-tunnel system, and a detailed blow-up in the tunnel, are presented as function of space and time for a flow tube with time independent cross section $A(\xi_1)$. c,d) Pressure is now presented in a flow tube with time dependent cross section $A(\xi_1, t)$. Around the dike-drift intersection the cross section is growing from 0.589 m^2 to 23.56 m^2 in one minute, whereafter it is fixed to the value in (a,b) . This is a simplified representation of the growth of the dike opening after breakthrough. The initial hydrostatic pressure in (a,b) is higher than in (c,d) at the dike-drift intersection 17.5 MPa and 12.5 MPa , respectively; and the friction in (c,d) is a factor 10 lower than in (a,b). At $\xi_1 = 0$ the magma enters or leaves the domain, and at $\xi_1 \approx 1000 \text{ m}$ is the closed right end of the tunnel.

The approximate mathematical model therefore enables us to estimate the filling speed of the tunnels. Moreover, the pressure at the end of the tunnel and the final pressure are higher than the pressure required to fracture the rock (about 5 MPa). A preliminary conclusion could thus be that new fractures or pathways to the surface of YM can form, which could possibly increase the amount of pollution flowing into the atmosphere.

At the same time, these preliminary insights raise many unanswered questions: How can we model the complex multiphase magma-water mixture more realistically? How does the rock of the magma dike react to the changing pressure in and around the tunnel and vice versa? It is fascinating that to answer these questions, which appear essential for the YM

project, we first have to answer fundamental questions in multiphase and rock mechanics.

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A Flow tube model

Consider a two-phase magma mixture of molten rock with density σ , liquid H_2O , and water vapor bubbles. The total mass fraction of water in the magma is n_0 (weight percent *wt%*) of which $n_s(p; \dots)$ is dissolved in the magma as liquid and $n_e(p; \dots)$ is exsolved in the bubbles. We have explicitly denoted the dependence of the mass fractions on the pressure, but in general they depend on a large number of variables. The density of water vapor is ρ_g . The total void or bubble fraction of the mixture is α . Consider an element of mass $\Delta M = \rho \Delta V$ of the two-phase mixture. It consist of the mass of the exsolved water vapor $\Delta M_g = \rho_g \Delta V_g$ which is the mass fraction n_e of ΔM . Furthermore, $\Delta V_g = \alpha \Delta V$. Hence, we obtain

$$\Delta M_g = \rho_g \alpha \Delta V = n_e \rho \Delta V \iff \alpha = n_e \rho / \rho_g. \quad (\text{A.1})$$

In addition, ΔM consists of the mass of the bubble void fraction $\alpha \Delta V$ and the remaining liquid volume $(1 - \alpha) \Delta V$

$$\Delta M = \rho \Delta V = (\alpha \rho_g + (1 - \alpha) \sigma) \Delta V. \quad (\text{A.2})$$

Combining these relations and using the ideal gas law $\rho_g = p/(RT)$, we find

$$\alpha = \frac{n_e / \rho_g}{n_e / \rho_g + (1 - n_e) / \sigma} \quad \text{and} \quad (\text{A.3})$$

$$\rho = \left(\frac{n_e R_v T}{p} + \frac{1 - n_e}{\sigma} \right)^{-1}, \quad (\text{A.4})$$

provided there are bubbles, i.e., when $n_e > 0$.

The next question is on how we determine n_0 and n_e is an open question. We therefore introduce the simplifying assumption that the two phases are in equilibrium and resort to laboratory measurements of magma under isothermal conditions. Petrologists have measured that the weight fraction of exsolved volatiles is given by [6]

$$n_e \equiv n(p, n_0) = n_0 - n_{s_H} \equiv n_0 - s_H p^\beta. \quad (\text{A.5})$$

In general, the total mass fraction of water, n_0 , may vary from fluid parcel to fluid parcel. At the critical pressure $p_c = (n_0/s_H)^{1/\beta}$, we then find $n(p_c, n_0) = 0$. For $p > p_c$ we assume the sound speed $c_s^2 = \partial p / \rho$ is constant. Hence, we find the following equation of state

$$\rho(p, n_0) = \begin{cases} \left(\frac{n(p, n_0) R_v T}{p} + \frac{1-n(p, n_0)}{\sigma} \right)^{-1}, & p < p_c \\ \sigma + c_s^{-2} (p - p_c), & p \geq p_c \end{cases}. \quad (\text{A.6})$$

Note that the sound speed is discontinuous at p_c .

A one-dimensional system, obtained by averaging over cross sections of the flow tube (see Fig. 2) and by neglecting flow in the cross-sectional planes, is derived by considering mass and momentum density conservation in a control volume $A(\xi_1, t) d\xi_1$. Momentum density is defined as $\rho A u$ with $u = u_1$ the ξ_1 -component of the velocity. Pressure in the ξ_1 -direction acts both on the slices at ξ_1 and $\xi_1 + \Delta\xi_1$, giving a contribution $\sim -\partial(p A)/\partial\xi_1$, and on the flow tube walls between these slices, giving a contribution $\sim p \partial A / \partial\xi_1$. The gravitational force is a volume force. The resulting equations of motion (following, e.g., [4], §8.1) are

$$\begin{aligned} \frac{\partial(\rho A u)}{\partial t} + \frac{\partial}{\partial\xi_1} \left(\rho A u^2 + p A \right) &= -A F_1 + p \frac{\partial A}{\partial\xi_1} - \rho g A \frac{\partial z}{\partial\xi_1}, \\ \frac{\partial(\rho A)}{\partial t} + \frac{\partial}{\partial\xi_1} \left(\rho A u \right) &= 0, \\ \frac{\partial(\rho A n_0)}{\partial t} + \frac{\partial}{\partial\xi_1} \left(\rho A n_0 u \right) &= 0 \end{aligned} \quad (\text{A.7})$$

with z the vertical coordinate, and $-F_1$ the parameterized forcing and/or dissipation. Note that we have included the advection of n_0 by the fluid in the absence of sources and sinks of water in the flow tube.

The determination of the frictional effects, which generally include a parameterization of the Reynold's terms resulting after averaging the three-dimensional flow over the cross section A , is an unresolved problem. We therefore approximate the frictional term F_1 as the sum of a term linear in u for low Reynolds number flow based on a viscous, Poiseuille flow analysis in a dike; and, a quadratic term proportional to $u |u|$ for high Reynolds number flow as a simple turbulent closure. Following the empirical parameterization of a high-viscosity magmatic foam in [7], we take

$$F_1 = \frac{3 \mu_0 e^{(5-100 n_0)} (1 - \phi)^{-5/2}}{2 L_e^2} u, \quad (\text{A.8})$$

where L_e is a typical length scale, for example the width of the dike or the radius of drift or conduit, and $\mu_0 = 10 - 100 \text{ Pa s}$. Relation (A.8) has validity when the void fraction of the mixture, ϕ , given by

$$\phi = \frac{1}{1 + [1 - n(p)] p / [n(p) R_v T \sigma]} \quad (\text{A.9})$$

remains below the fragmentation threshold $\phi < \alpha$, where $\alpha = 0.7 - 0.9$ [8]. As ϕ evolves through this regime, the gas becomes the continuous phase and frictional forces diminish. When $\phi > \alpha$, a simple parameterization for turbulent flow in a pipe was proposed in [9]:

$$F_1 = 0.0025 \frac{\rho |u| u}{L_e}. \quad (\text{A.10})$$

The hydraulic radius L_e refers to the geometry of the cross section. In the inviscid limit when $F_1 = 0$ the shape of the cross sections is arbitrary while their area is specified, but when the frictional terms described above are added the area shape is specified furthermore through a characteristic cross-sectional width L_e . In the dike-drift system L_e is five times smaller in the dike than in the drift; friction in the dike is thus five to 25 times larger than in the drift.

The one-dimensional, averaged compressible equations of motion for air (following [4], §8.1) are:

$$\begin{aligned}\frac{\partial(\rho_a A u_a)}{\partial t} + \frac{\partial}{\partial \xi_1} \left(\rho_a A u_a^2 + p_a A \right) &= -A F_a + p_a \frac{\partial A}{\partial \xi_1} - \rho_a g A \frac{\partial z(\xi_1)}{\partial \xi_1}, \\ \frac{\partial(\rho_a A)}{\partial t} + \frac{\partial}{\partial \xi_1} \left(\rho_a A u_a \right) &= 0, \\ \frac{\partial \Omega}{\partial t} + \frac{\partial}{\partial \xi_1} \left((\Omega + p_a A) u_a \right) &= -p \frac{\partial A}{\partial t} - u_a A F_a,\end{aligned}\tag{A.11}$$

in which subscripts “a” distinguish variables in air from ones in the basaltic fluid, and in which

$$\Omega = \rho_a A \left(\frac{1}{2} u_a^2 + e + g z \right)\tag{A.12}$$

is the energy density of air with internal energy e . Air is modeled as an ideal gas with $p_a = \kappa(s) \rho_a^\gamma = (\gamma - 1) \rho_a e$, where $\kappa = \kappa(s)$ is a function of entropy s and $\gamma = c_p/c_v = 1.4$ is the ratio of specific heats at constant pressure and volume, respectively. Viscous forces in air will be ignored, that is, $F_a = 0$.

The basaltic fluid and air are separated by an interface. In the one-dimensional flow-tube model, the interface between the basaltic fluid on one side and air on the other is marked by a fluid parcel at position $\xi_1 = \xi_i(t)$. The dynamics of this parcel is governed by

$$\frac{d\xi_i}{dt} = u_i, \quad \text{and} \quad \frac{du_i}{dt} = -\frac{1}{\rho} \frac{\partial p}{\partial \xi} \Big|_{\xi_i(t)} = -\frac{1}{\rho_a} \frac{\partial p_a}{\partial \xi_1} \Big|_{\xi_i(t)},\tag{A.13}$$

where velocity $u_i(t) = u(\xi = \xi_i(t), t)$ and pressure are continuous across the interface, while density is generally not. Since we focus on high-speed inertial effects, diffusion of heat across the interface is neglected. The boundary conditions for the one-dimensional flow tube model are in- or outflow at the beginning of the flow tube or dike at depth in the Earth, and $u = 0$ at the end wall in the tunnel. In- or outflow conditions are either implemented by simply extrapolating interior values, or by using an approximate approach based on the Riemann invariants for the frictionless situation with a locally constant cross section A .

B Hydrostatic relation

The hydrostatic relation $\partial p / \partial \xi_1 = -\rho g \partial z / \partial \xi_1$ can be integrated by using (A.6) to get an implicit relation between pressure and vertical coordinate z . This is most straightforward when n_0 is constant. These pressure values are used to initialize the model in the dike at rest. The initial condition of the air in the tunnel is one of rest, room temperature and atmospheric pressure. Assuming we know the pressure p_m and coordinate z_m at the

dike-drift intersection we find for constant n_0

$$n_0 R_v T \ln(p_m/p) - \frac{s_H}{\beta} (p_m^\beta - p^\beta) R_v T + \frac{(1-n_0)(p_m-p)}{\sigma} + \frac{s_H}{\beta+1} \left(\frac{p_m^{\beta+1} - p^{\beta+1}}{\sigma} \right) = g(z - z_m) \quad \text{for } p < p_c \quad (\text{B.1})$$

and

$$n_0 R_v T \ln(p_m/p_c) - \frac{s_H}{\beta} (p_m^\beta - p_c^\beta) R_v T + \frac{(1-n_0)(p_m-p_c)}{\sigma} + \frac{s_H}{\beta+1} \left(\frac{p_m^{\beta+1} - p_c^{\beta+1}}{\sigma} \right) + c_s^2 \ln \left(\frac{\sigma}{\sigma + (p - p_c)/c_s^2} \right) = g(z - z_m) \quad \text{for } p \geq p_c. \quad (\text{B.2})$$

C Geometry of flow tube model

We define the geometric functions $z(\xi)$ and $A(\xi, t)$ in the flow tube model (A.7), used in the presented simulations, in this appendix in addition to the hydraulic radius of the flow tube used in the frictional parameterizations.

The vertical coordinate as function of the flow tube coordinate ξ is given by

$$z(\xi) = \begin{cases} \xi, & 0 \leq \xi < L_{dike} + \frac{1}{2}(d-w) \\ L_{dike} + \frac{1}{2}(d-w) + \frac{1}{2}w \sin \{2[\xi - L_{dike} + \frac{1}{2}(w-d)]/w\}, & L_{dike} + \frac{1}{2}(d-w) < \xi < L_{dike} + \frac{1}{2}d + (\frac{\pi}{4} - \frac{1}{2})w \\ L_{dike} + \frac{1}{2}(d-w) + \frac{\pi}{4}w, & L_{dike} + \frac{1}{2}d + (\frac{\pi}{4} - \frac{1}{2})w < \xi < L_{end} \end{cases} \quad (\text{C.1})$$

with L_{end} the length of the flow tube, L_{dike} the length of dike considered, w the width of the dike, and d the diameter of the tunnel. $L_{end} = L_{dike} + \pi w/4 + d/2 - w/2 + L_{tunnel}$.

The effective cross-sectional area $A(\xi)$ for the fixed tunnel is given by

$$A(\xi) = \begin{cases} \frac{1}{2} A_{dike}, & 0 \leq \xi < L_{dike} - L_2 \\ \frac{1}{2} A_{dike} + \frac{1}{2} (\pi d w/2 - A_{dike}) \times (1 - \cos \{ \pi [\xi - L_{dike} + L_2]/L_2 \}), & L_{dike} - L_2 \leq \xi \leq L_{dike} \\ \pi d w/2 + (A_{tunnel} - \pi d w/2) \times (1 - \cos \{ \pi (\xi - L_{dike})/L_e \}), & L_{dike} < \xi < L_{dike} + L_e \\ A_{tunnel}, & L_{dike} + L_e \leq \xi < L_{end} \end{cases} \quad (\text{C.2})$$

with $L_e = m d + (\pi/4 - 1/2) w - d/2$ for some integer $m \in [1, 20]$, and $L_2 = L/2$ half of the spacing L between drifts. Since half the flow is entering the left and right part of the tunnel, respectively, the effective cross section is divided by two as well as the width w .

The hydraulic radius L_e used in the frictional parameterizations (A.8)–(A.10) is

$$L_e = R_h(\xi) = \begin{cases} \sqrt{A_{tunnel}/\pi} = d/2, & L_{end} - L_{tunnel} < \xi < L_{end} \\ A(\xi)/\pi, & L_{end} - L_{tunnel} - L_{arc} < \xi < L_{end} - L_{tunnel} \\ w, & 0 < \xi < L_{end} - L_{tunnel} - L_{arc} \end{cases} \quad (\text{C.3})$$

with $L_{arc} = (\pi/4 - 1/2) w + d/2$.

Parameter	simulation a	simulation b
T	1350 K	1350 K
σ	2500 kg/m^3	2500 kg/m^3
R_v	462 $J\,kg^{-1}\,K^{-1}$	462 $J\,kg^{-1}\,K^{-1}$
β	0.5	0.5
s_H	$3 \times 10^{-6}\,Pa^{-1/2}$	$3 \times 10^{-6}\,Pa^{-1/2}$
n_0	2 wt%	2 wt%
c_s	1400 – 2000 m/s	1400 – 2000 m/s
$P_{dike\,tip}(t = 0)$	17.5 MPa	12.5 MPa
L_{drift}	200 m	200 m
A_{dike}	120 m^2	120 m^2
A_{tunnel}	19.64 m^2	19.64 m^2
L_{dike}	800 m	800 m
L	80 m	80 m
α	0.7	0.7
T_{end}	0	60 s
μ_0	10 $Pa\,s$	100 $Pa\,s$
w	1.5 m	1.5 m
d	5 m	5 m
integer m in L_e	4	10

Table 1: Parameter values used in the two simulations of figure 3.

The effective cross-sectional area $A(\xi, t)$ for a growing tunnel is given by

$$A(\xi, t) = \begin{cases} \frac{1}{2} A_{dike}, & 0 \leq \xi < L_{dike} - L_2 \\ \frac{1}{2} A_{dike} + \frac{1}{2} \left[\left(\frac{1}{40} + \frac{39}{40} \frac{t}{T_{end}} \right) \pi d w / 2 - A_{dike} \right] \times \\ \quad (1 - \cos \{ \pi [\xi - L_{dike} + L_{drift}] / L_{drift} \}), & L_{dike} - L_2 \leq \xi \leq L_{dike} \\ \pi d w + (A_{tunnel} - \pi d w / 2) \times \\ \quad (1 - \cos \{ \pi (\xi - L_{dike}) / L_e \}), & L_{dike} < \xi < L_{dike} + L_e \\ A_{tunnel}, & L_{dike} + L_e \leq \xi < L_{end} \end{cases} \quad (C.4)$$

for $t < T_{end}$. The cross-sectional area is continuous and has reached its fixed value at $t = T_{end}$.

D Simulations

The equations of motion are discretized using shock capturing second- and third order Local Lax Friedrichs and Essentially Non-Oscillatory numerical schemes and tested extensively, see [10].

For the presented simulations we have taken n_0 is constant. In the two presented simulations all pressures remain below p_c . All the remaining values for the parameters given in the relations the appendices and used in the two simulations are summarized in table 1.

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